YANQING LIU

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RESEARCH FOCUS

My research interests lie at the intersection of human-AI interaction, multimodal sensing and adaptive systems. I study how interaction, conversational and physiological data can reveal users' momentary cognitive and emotional states, and how AI systems can adapt to them to better support people's learning, work, decision-making and well-being.

EDUCATION

University of New South Wales **Sydney, Australia | 2025 – 2026**

Graduate Certificate in Information Technology (exit award from the Master of IT) | GPA: 91.625/100, High Distinction

Coursework: *Interactive Design*, *Data Structure and Algorithm*, *Web Front-End Programming*

Columbia University **New York, USA | 2021 – 2023**

M.S. in Applied Analytics | GPA: 4.132/4.3

Coursework: *Research Design*, *Applied Text and Natural Language Analytics*, *Storytelling with Data*

Macau University of Science and Technology **Macau, China | 2016 – 2020**

Bachelor of Applied Economics | Major GPA: 3.86/4.0 | Rank: 2/882, School of Business

Coursework: *Advanced Mathematics*, *Linear Algebra*, *Statistics*, *Behavioral Economics*, *Game Theory*

PUBLICATIONS & PRESENTATIONS

Liu, Y. "From Sensing to Adaptation: A Scoping Review of Physiological Indicators of Learner States for Adaptive Learning Support." *HCI International 2027 (HCII 2027)*, Springer. [Accepted; full paper in preparation]

Liu, Y. "Mindful-Scroll: Linking Interaction Logs to Self-Reported Mental Health and Well-Being, and In-App Interventions." *Digital Mental Health International Conference 2026*, Hong Kong, China, June 2026. [Presentation]

Liu, Y., Zhou, W., Zhu, Y., Zhao, X., & Jin, Y. "Negotiating Reflective Demand: LLM-Mediated Reflection on Everyday Short-Form Video Use." *CHI 2027*. [Under Review]

Liu, Y., Zhu, Y., & Jin, Y. "Exploring Candidate Signals of Perceived Stress and Anxiety in Short-Form Video Use: Scrolling Logs and Use-Specific Appraisals." In *DMH Conference Book: Maximizing the Real-World Potential of Digital Mental Health*. Springer Nature. [Under Review]

Liu, Y., Zhu, Y., & Jin, Y. "Beyond Social-Media Posts: Conversational Reflection Signals of Daily Perceived Stress and Anxiety in Short-Form Video Use." [Manuscript in revision for journal resubmission]

Wang, Y., **Liu, Y.**, & Ray, L. C. "'Like Fire Alarm Promotes Waking Up': How Humor Can Affect User Engagement with Misinformation on Social Media." *CHI 2027 Late-Breaking Work*. [In Preparation]

Deng, W., **Liu, Y.**, Chen, Y., Zhang, H., Li, Y., Mi, H., & Yu, C. "LLM-Based Simulation to Support Student Academic and Life Supporters (Fudaoyuan) at Chinese Universities." *CHI 2027 Late-Breaking Work*. [In Preparation]

RESEARCH EXPERIENCE

Research Collaborator **Singapore Management University** **09/2026 – Present**

Advisor: Prof. Fiona Fui-Hoon Nah | School of Computing and Information Systems

Lead a scoping review of physiological learner-state indicators for adaptive learning (accepted at HCII 2027).

- Designed the protocol (JBI, PRISMA-ScR) and an iteratively tested Scopus search (5,211 unique records).
- Conducted a 200-record screening and data-charting pilot to test the eligibility criteria and coding scheme.

Research Fellow (Full-Time) **Duke Kunshan University** **02/2026 – Present**

Advisor: Prof. Yucheng Jin | Hybrid Interactive Intelligence (HII) Lab

Lead Mindful Scroll, a mixed-methods project on short-form video use, well-being and LLM-mediated reflection.

- Ran a pilot survey (n = 32), identifying a four-trace behavioral usage cluster linked to stress and anxiety.
- Independently developed *Mindful Scroll*, a React-based, Dockerized short-form video platform, and ran a 14-day field study with 41 participants, collecting 44,420 video views and 113,233 log events.
- Designed an analysis pipeline combining GPT-assisted seeded topic modeling, predictive modeling, and longitudinal analyses to investigate how usage and conversation patterns relate to stress and anxiety.
- Conducted 12 participant interviews to examine the perceived benefits and burdens of reflection, informing the design of interactions that give users control over reflection.

Summer Research Intern **City University of Hong Kong** **06/2026 – 08/2026**

Advisors: Prof. Ray LC and Prof. Fiona Fui-Hoon Nah (co-advisor) | Studio for Narrative Spaces

Developed an experimental interface to study how comment conditions shape responses to misinformation.

- Led front-end development of a simulated social media feed for a between-subjects experiment, with randomized conditions and logging of share, repost, report and skip decisions.
- Led the quantitative analysis of 1,200 post-level decisions from 60 participants using GEE models.
- Co-authored a CHI 2027 Late-Breaking Work manuscript.

Research Collaborator **Tsinghua University** **09/2025 – Present**

Advisor: Dr. Jie Cai | Pervasive HCI Group

Analyzed conversations in CoSim, an LLM simulation for training university student academic and life supporters.

- Built a trajectory-then-thematic pipeline across 10 supporters and 19 sessions to trace emotion, self-disclosure, and resistance across turns and contrast facilitative versus hindering sessions.
- Identified communication patterns, surfacing supporters' blind spots in emotionally sensitive conversations and informing simulation-based feedback design.
- Contributed to the mixed-methods analysis and co-authored a CHI 2027 Late-Breaking Work manuscript.

Course Research Projects **Columbia University** **2022**

Instructors: Prof. Kitty Kay Chan and Prof. Cesar Koirala | Industry Partner: Sandler Training Company

- Analyzed 200,000+ Yelp reviews with text mining, sentiment analysis and decision trees to identify pandemic-related themes and their links to ratings.
- Analyzed 479 client surveys with clustering, classification and random forest models, segmenting clients by revenue, growth and sentiment to identify high-value groups.

WORK EXPERIENCE

Frontend Developer Intern **Beijing Puppy Agent Technology Co., Ltd.** **08/2024 – 12/2024**

- Developed an enterprise RAG-based workflow platform, designing a hybrid ReactFlow state management architecture, cutting workflow creation time from 2 hours to 30 minutes.
- Built a node framework with automated edge generation and integrated Monaco Editor for rich-text editing.

SKILLS

Technical: Python • R • SQL • JavaScript • React • Docker • LLM APIs • Figma • MATLAB • Stata

Methods: Mixed methods • Thematic analysis • Topic modeling • Predictive modeling • A/B experiments

Languages: Mandarin (native) • English (fluent) | TOEFL iBT 104 • GMAT 690

AWARDS & HONORS

Australia's Global University Award, UNSW	2025
Dean's Scholarship, MUST	2019
Luso International Banking Ltd. Scholarship, MUST	2019
Nam Kwong (Group) Company Limited Scholarship, MUST	2018